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Multimodal Techniques for Financial Stock Forecasting

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Abstract

Accurate financial forecasting remains a fundamental challenge in quantitative finance, where price histories alone fail to capture the full spectrum of signals that drive market behavior. It requires integrating heterogeneous signals ranging from raw price histories to unstructured financial news and structured corporate filings. Recent multimodal datasets such as FinMultiTime enable cross-modal financial analysis, but the comparative performance of different model architectures on such data remains underexplored. We present a benchmarking study on S&P 500 constituents using three complementary modalities: historical price data, financial news, and SEC XBRL filings. We compare unimodal baselines against pretrained foundation models and multimodal fusion architectures. Our central finding is that the choice of fusion mechanism dominates the choice of temporal encoder: cross-attention and encoder-decoder fusion between a temporal encoder and FinBERT improve on unimodal baselines, while concatenation-based late fusion underperforms. These findings provide practical guidance for designing multimodal financial forecasting systems at scale.

Keywords: multimodal forecasting; stock price prediction; financial news; FinBERT; time-series modeling; S&P 500.

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1 Introduction

Financial time-series forecasting has been a long-standing challenge at the intersection of economics, statistics, and machine learning. Traditional approaches rely mostly on historical price data and often fail to capture the abrupt sentiment dynamics of real-world financial markets. This limitation is not merely technical: asset prices aggregate beliefs formed across qualitative and quantitative channels simultaneously, so a system restricted to price histories discards information that market participants actively use.

At the same time, financial decision-making inherently depends on heterogeneous information sources, including news, corporate disclosures, market signals and much more. Recent advances in deep learning and large language models (LLMs) have enabled the extraction of meaningful signals from unstructured data such as financial news, while structured sources like SEC filings provide reliable indicators of firm fundamentals. This shift reflects a broader trend toward multimodal learning, where integrating diverse data sources improves predictive performance over unimodal price-only baselines [13].

Datasets such as FinMultiTime [16] have further accelerated this direction by providing multiple modalities — including text, reports, and time series — in one unified dataset. However, existing multimodal studies tend to introduce new data sources and new architectures simultaneously, making it impossible to attribute performance gains to either factor. This study addresses that gap by holding the evaluation protocol and prediction target fixed across all architectures, isolating the contribution of each modality

2 Problem Statement

This project investigates the following research question: *How do progressively richer input modalities, and the mechanisms by which they are fused, affect the accuracy of short-horizon stock price forecasting on S&P 500 equities?*

Formally, let $\mathbf{x} = \{x_{t-T+1}, \ldots, x_t\}$ denote a lookback window of historical close prices for a given stock, where T is the input sequence length and x_t is the closing price at trading day t. In addition to this price sequence, each sample may include aligned financial news text and periodic financial statement data representing sentiment-driven and fundamental information, respectively.

The forecasting objective is to predict the next H trading-day close prices,

$$\mathbf{y} = \{x_{t+1}, \ldots, x_{t+H}\},$$

where H is the prediction horizon. Across the experiments described in this report, $H = 5$ trading days and $T \in \{30, 60, 90\}$ trading days, depending on the architectural variant under evaluation.

The central challenge is therefore not only to encode and align different inputs, but also to determine whether non-price modalities can be fused with price histories in a way that improves predictive accuracy relative to unimodal baselines. The output target is unified across architectures with

appropriate inverse transformations applied for evaluation. This framing allows us to compare how different architectures use complementary information from market behavior, event-driven news signals, and lower-frequency financial disclosures under a consistent short-horizon forecasting task.

3 Related Work

Stock price forecasting has always been talked about in the machine learning community, with approaches evolving from classical statistical models toward increasingly expressive deep learning architectures. Long Short-Term Memory networks [6] established the foundational paradigm for sequential equity prediction, with Fischer and Krauss [4] demonstrating their superiority over random forests on S&P 500 data. A broad benchmarking survey by Ozbayoglu et al. [13] later concluded that no single architecture dominates across market conditions and that hybrid, multimodal approaches tend to outperform unimodal price-only baselines, a motivating finding for this study.

Transformer-based architectures substantially advanced the state of the art in sequence modelling. The Temporal Fusion Transformer (TFT) [9] introduced attention-based multi-horizon forecasting with explicit handling of static covariates, known future inputs, and observed exogenous time series, alongside gating mechanisms that suppress irrelevant components. Its covariate-aware design makes it directly applicable to multimodal financial inputs. PatchTST [11] proposed segmenting time series into subseries-level patches as input tokens, combined with channel-independent processing, reducing attention complexity quadratically while preserving local temporal semantics, a particularly appealing property for multi-day equity pattern recognition. Concurrently, DLinear [17] showed that simple linear decomposition models could rival complex Transformers on standard benchmarks, prompting renewed interest in efficient architectures.

TSMixer [3] responded to this efficiency imperative with an all-MLP architecture that interleaves time-mixing and feature-mixing layers to capture temporal dynamics and cross-variate correlations without attention overhead. On the large-scale M5 benchmark, TSMixer outperformed all alternatives while remaining simple to implement, and its extended formulation natively supports static and dynamic auxiliary covariates, making it well-suited as a backbone for modality ablation experiments. On the foundation model front, Chronos [2] adapted T5-family language model architectures to time series via quantisation-based tokenisation and pre-training on 42 diverse datasets. While Chronos demonstrated strong zero-shot transfer in general domains, Sakib et al. [14] found it underperforms on financial equity data due to non-stationarity and abrupt regime changes, further motivating domain-specific signal integration.

On the textual side, FinBERT [7] fine-tuned BERT on financial news and earnings transcripts to produce contextualised sentiment representations better calibrated to financial language than general-purpose models or lexicon-based approaches. Empirical studies demonstrate that FinBERT embeddings improve directional accuracy in next-day return prediction, particularly around information-rich events such as earnings announcements. Integrating such representations with price history has been a central theme in recent multimodal stock prediction work. He and Gu [5] proposed a multi-modal attention network that weights textual sources by poster credibility before fusing semantic and numeric features, achieving 61.20% directional accuracy and significantly improved simulated trading profits over unimodal baselines. A 2023 systematic comparison of fusion strategies

found that late fusion offers training stability while attention-based cross-modal fusion yields the highest accuracy when sufficient data is available. Most recently, DASF-Net [12] combined FinBERT sentiment aggregated over an empirically optimised three-day window with graph diffusion over industry- and fundamental-similarity networks, fused via multi-head attention, achieving up to 91.6% relative MSE reduction on S&P 500 large-cap stocks at one-to-three day horizons, the most directly comparable setup to this study.

Feature engineering for financial prediction has been surveyed extensively by Shah et al. [15], who reviewed over 150 papers and found that ensemble feature selection methods combining filter, wrapper, and embedded strategies produce the most robust technical input sets. Standard features include momentum indicators (RSI, MACD), trend measures (EMAs, Bollinger Bands), volume signals (OBV, VWAP), and volatility proxies (ATR, historical volatility), complemented in richer regimes by fundamental accounting ratios, macroeconomic covariates (VIX, risk-free rate), and sector-level signals. These structured features operationalise market-wide co-movement that price history alone cannot capture without impractically long look-back windows.

With respect to loss functions, Mean Squared Error remains the standard training objective but is sensitive to the fat-tailed return distributions characteristic of equity markets. Huber loss [8], combining MSE for small errors with MAE for large deviations via a tunable threshold, provides a robust alternative that prevents extreme return events from dominating the gradient signal and has been adopted in several recent financial forecasting systems including DASF-Net [12].

Taken together, the existing literature establishes the individual value of patch-based transformers, all-MLP mixing architectures, FinBERT sentiment features, and structured financial covariates for equity forecasting, but rarely isolates the marginal contribution of each modality in a controlled ablation. Most multimodal studies introduce architectural novelty alongside new data modalities, making it impossible to attribute performance gains to either. This study addresses that gap directly: by holding the forecasting backbone constant across three progressively richer modality configurations — price time series alone, price time series with news sentiment, and price time series with both news sentiment and structured financial features — we isolate the informational contribution of each data source independently of architectural effects.

4 Methodology

4.1 Dataset

All experiments in this project are conducted on the FinMultiTime Dataset [16], a large-scale dataset covering multiple modalities of information for stock forecasting. The original dataset spans the period from 2000 to 2025 and integrates four modalities—financial news text, financial reports, technical chart images, and stock price time series—across both U.S. (S&P 500) and Chinese (HS300) markets. While the stock price time-series data covers the full temporal range, the remaining modalities begin later, with coverage starting from 2009. To ensure a realistic evaluation setting and avoid look-ahead bias, the data is partitioned using a temporal split, with 70% used for training, 15% for validation, and 15% for testing.

Dataset Type	Size	Tickers Count	Frequency
Semi-Annual Reports	84.04 GB	2,676	Semi-Annual
Stock Timeseries	1.83 GB	4,213	Daily
Daily Text	14.1 GB	4,694	Daily
Total (All)	108.64 GB	—	—

Table 1: Summary of the FinMultiTime dataset used in this study.

In this work, we construct a reduced and task-specific version of the dataset by excluding image data and HS300 (Chinese market) samples, focusing exclusively on S&P 500 constituents.

4.1.1 Stock Price Data

Daily stock price data including open, high, low, close, and volume (OHLCV) are sourced for all available S&P 500 tickers. These series are normalized per stock to ensure stationarity and comparability across assets.

4.1.2 Financial News Data

Financial news articles associated with each stock include date, headlines, full text body, and source. Articles are summarized and processed to extract sentiment signals, providing qualitative context for price movements.

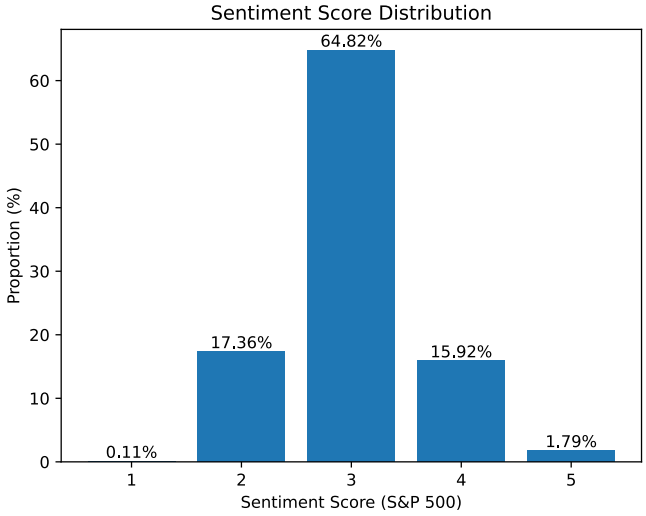


Figure 1: Distribution of article-level sentiment labels in the news subset.

For the S&P 500 subset, sentiment labels were represented on a five-point ordinal scale: 1 denoted negative, 2 denoted somewhat negative, 3 denoted neutral, 4 denoted somewhat positive, and 5 denoted positive. Models that employ FinBERT directly bypass this ordinal encoding and instead operate on the full 768-dimensional contextualized [CLS] embedding, which carries richer polarity, intensity, and domain-specific information than the scalar label.

4.1.3 Financial Reports & Earnings Data

Structured financial indicators were derived from company filings, including key variables related to profitability, liquidity, and capital structure, such as cash flow, equity, and earnings. These variables were aligned with reporting periods and forward-filled to match trading days so that fundamental signals could be synchronized with daily market observations. This data was sourced from the SEC EDGAR API, which exposes a nested dictionary structure requiring careful extraction logic during preprocessing to isolate the relevant financial fields from each filing.

4.1.4 Sector Relational Data

Each ticker was associated with its GICS sector and used to construct sector-based relational edges in graph models, enabling the pipeline to represent shared industry exposure and sector-level movement.

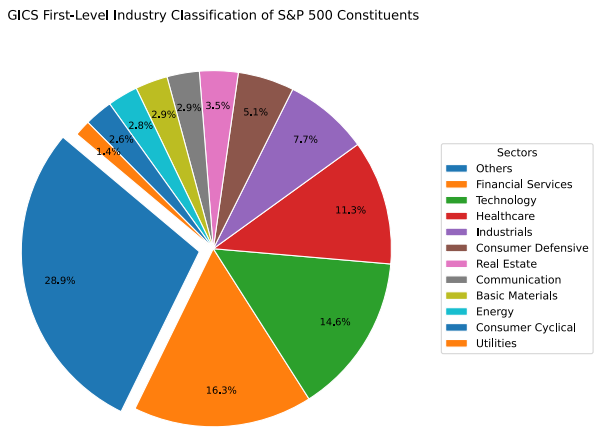


Figure 2: Distribution of stocks across GICS sectors in the relational metadata.

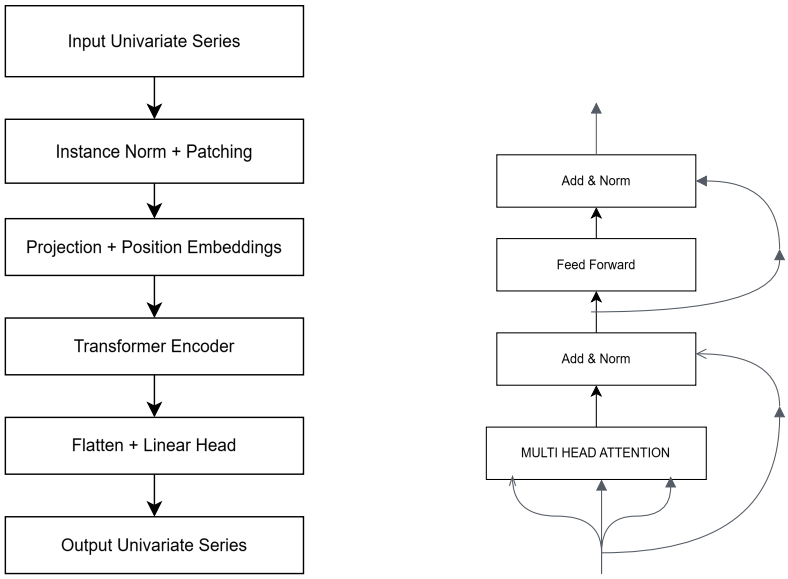
4.2 Model Architectures

4.2.1 Unimodal Temporal Baselines

LSTM (Long Short-Term Memory). A two-layer stacked LSTM with hidden dimensions 128 and 64, trained on engineered OHLCV-derived features with a 60-day lookback window. The model uses a direct multi-step output head that produces all five future log returns $\{r_{t+1}, \ldots, r_{t+5}\}$ in a single forward pass, avoiding compounding autoregressive error. These log returns are accumulated onto the last observed close price x_t at inference. A composite loss combining Huber regression [8] ($\delta = 0.005$) with a directional binary cross-entropy term encourages both calibrated magnitude and sign prediction.

The feature input set combines log returns, rolling return statistics, RSI(14), ATR(14), MACD and MACD signal line and Bollinger bands.

PatchTST (Patch Time Series Transformer). We build on the original PatchTST implementation [11]. The architecture is a channel-independent Transformer that segments each input channel into overlapping patches of length 8 with stride 4, projects each patch to a 128-dimensional latent vector, and processes the resulting patch sequence through three Transformer encoder blocks with eight attention heads. RevIN is applied independently per channel at the input and its inverse is applied at the output, so that forecast is returned in close price domain $\{x_{t+1}, \ldots, x_{t+5}\}$. Training uses AdamW [10] with cosine annealing and Huber loss [8] ($\delta = 0.005$).



(a) PatchTST architecture (b) Transformer encoder

Figure 3: PatchTST architecture and its transformer encoder.

TSMixer (Time-Series Mixer). We build on the original TSMixer implementation [3], which treats rows as timesteps and columns as input variates and alternates between time-mixing and feature-mixing MLP blocks. The time-mixing MLP is shared across features, the feature-mixing MLP is shared across timesteps, and stacking N such mixer layers yields a parameter-efficient multivariate forecaster. Our implementation preserves this alternating mixer structure but introduces two task-specific modifications: an 8-dimensional learnable ticker embedding is appended to each sample to provide modest stock-specific specialization, and the original MSE objective is replaced with a combined Huber and quantile loss to improve robustness to heavy-tailed market moves.

As with PatchTST, RevIN normalization is applied at the input and inverted at the output so that the final forecast are expressed as close prices.

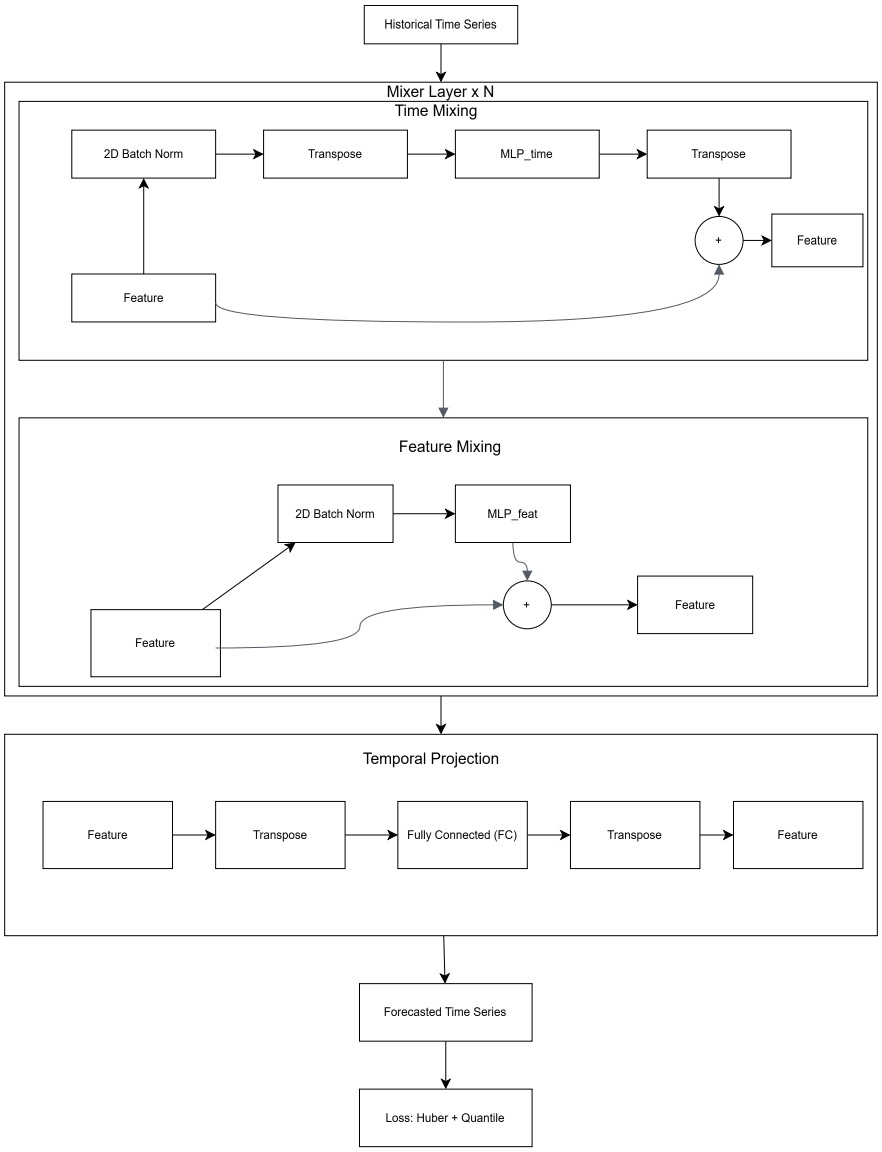


Figure 4: TSMixer architecture for multivariate time-series forecasting.

Chronos Foundation Models. We evaluate two pre-trained time-series foundation models: Chronos T5-Large [2], an encoder-decoder Transformer pre-trained on large time-series corpora via quantization-based tokenization, and Chronos-2 [1], its successor model. Both are evaluated on the test sample of stocks (sampling ratio 0.15, seed 42), with the final 5 trading days held out as ground truth. Both models take past time series data as input to predict several future prices. Instead of giving one exact value, they output a probability distribution over possible future close price ranges. The median of that distribution is taken as the point forecast, yielding a forecasted close price sequence $\{x_{t+1}, \ldots, x_{t+5}\}$ directly comparable to other architectures.

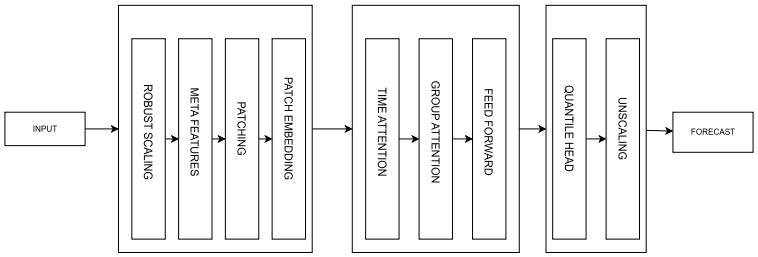


Figure 5: Chronos architecture diagram.

The Chronos pipeline (Figure 5) segments raw input data through a feature scaling stage, followed by input patching and patch embedding to produce compact temporal representations. These embeddings are processed through a Transformer block comprising time attention, group attention, and a feed-forward layer, enabling the model to capture both local and global temporal dependencies. A quantile head maps the learned representations to a distribution over future values, from which the median prediction is extracted for point-forecast evaluation.

TCN-Transformer. A hybrid architecture in which four stacked temporal convolutional blocks with dilated causal convolutions (dilation factors 1, 2, 4, 8) extract multi-scale local temporal features, which are then treated as temporal tokens by a two-layer Transformer encoder with eight attention heads. Sinusoidal positional encodings are injected before the Transformer stage. The model is trained on feature engineered representation (OHLCV plus Bollinger bands, RSI, and MACD components) with a 60-day lookback window. The prediction head outputs a 5-step vector of returns relative to the last observed close, which are accumulated at inference time to produce the close price sequence.

Temporal Fusion Transformer (TFT). The TFT [9] incorporates a Variable Selection Network (VSN) that learns per-variable importance weights using Gated Residual Networks (GRNs), an LSTM temporal encoder with gated skip connections, sinusoidal positional encoding, and interpretable multi-head attention with a shared value projection. The gated residual design throughout the TFT allows any component to act as a pass-through when its transformation is unhelpful, providing implicit regularization. Like the TCN-Transformer, the TFT is trained on the same feature representation with a 60-day lookback.

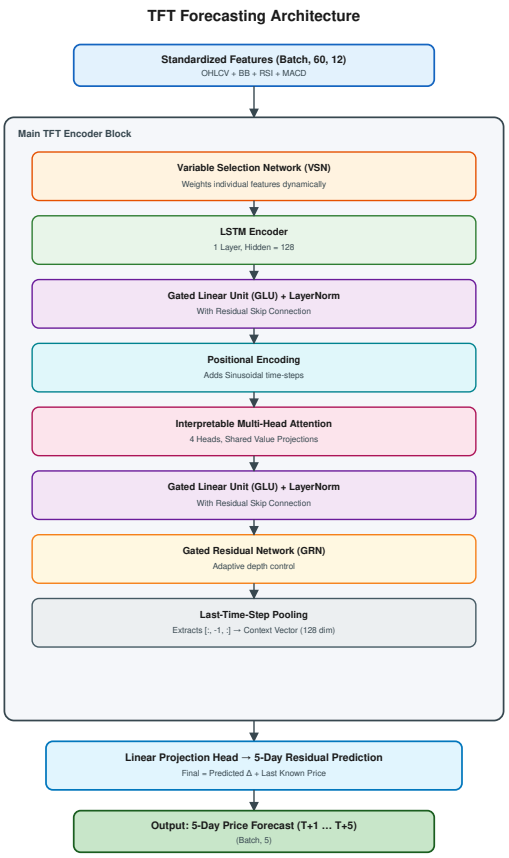


Figure 6: Temporal Fusion Transformer Architecture.

4.2.2 FinBERT

To isolate the predictive value of the news modality independently of any price signal, we evaluate a text-only forecasting model built on the pretrained FinBERT encoder [7]. For each sample, the most recent non-empty news headline within the input window is identified and encoded by FinBERT; the 768-dimensional `[CLS]` embedding is projected to a lower-dimensional news representation via a learned linear layer. This projected vector is concatenated with three numerical conditioning features — the last observed closing value and two normalization statistics derived during preprocessing — and passed through a two-layer MLP regression head that predicts the five-step close price sequence $\{x_{t+1}, \ldots, x_{t+5}\}$ in a single forward pass. No historical price sequence is provided as input; the model's predictive signal derives entirely from the textual modality and the scalar close anchor. The FinBERT encoder is kept trainable during optimization, and the model is trained with mean squared error loss using AdamW with separate learning rates for the pretrained backbone and the task-specific projection and regression layers. As expected, conditioning the forecast solely on the textual modality yields poor predictive performance: without access to the historical price sequence, the model fails to capture the temporal dynamics governing short-horizon price movement, and the resulting RMSE and MAE are comparable to those of a naïve random baseline, confirming that news sentiment alone — absent any structured price signal — carries insufficient information to support reliable multi-step forecasting.

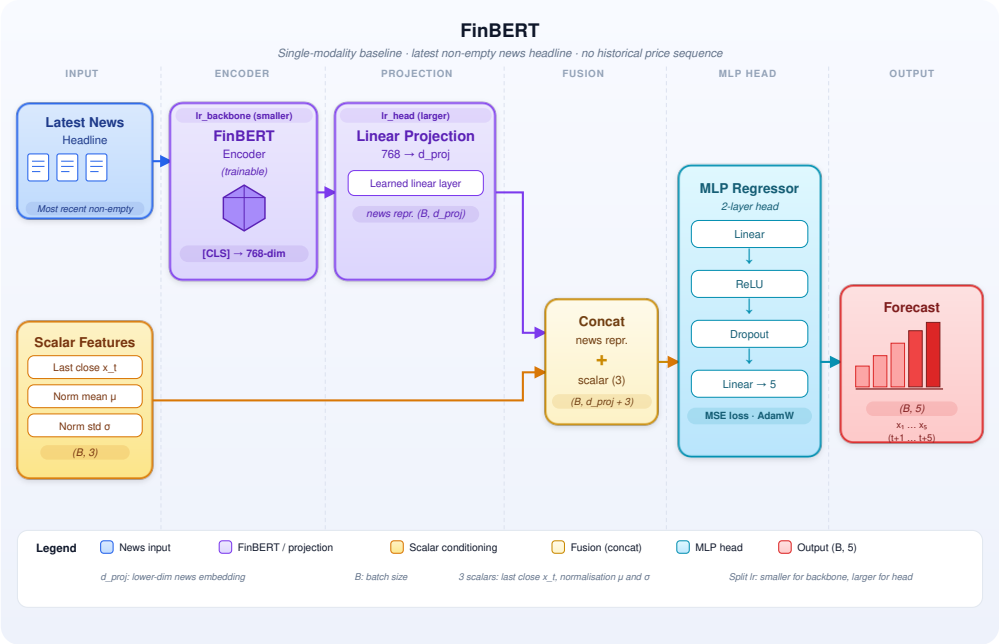


Figure 7: Finbert Baseline model architecture

4.2.3 Chronos-FinBERT

The ChronosFinBERT model adapts the encoder component of Chronos-T5-Base as a temporal feature extractor, pairing it with FinBERT for news-conditioned forecasting. A normalized close-price sequence of length $T \in \{30, 60, 90\}$ trading days is projected via a learned linear layer into the 512-dimensional Chronos encoder embedding space. Only the top four Transformer encoder blocks are fine-tuned; the remaining layers are frozen to preserve the temporal prior from pre-training. This yields a temporal representation matrix of shape $[B, T, 512]$.

The ablation over lookback lengths found $T = 60$ to produce the lowest validation RMSE; all results reported for ChronosFinBERT in Section 5 use this configuration. Simultaneously, the most recent non-empty news headline for each sample is encoded by FinBERT, and the [CLS] token embedding (dimension 768) is extracted; the top four FinBERT encoder layers are fine-tuned. Both the temporal and textual embeddings are independently projected to a common fusion dimension of 512 via learned linear layers. A multi-head cross-attention layer is then applied with the temporal embeddings as query and the sentiment embedding as key and value, allowing each temporal position to selectively condition on the news context. The cross-attended representation is passed through a two-layer LSTM, and a linear head projects the final hidden state to a 5-step forecast.

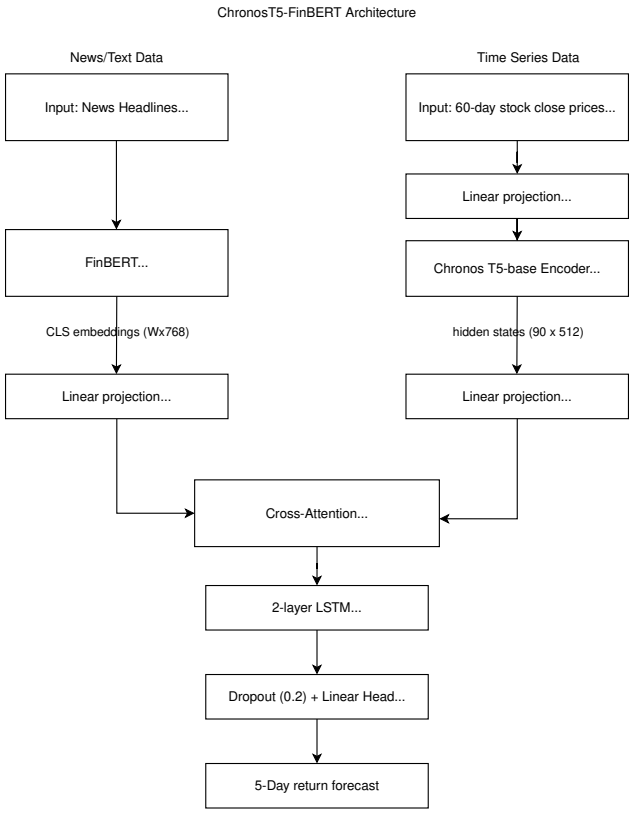


Figure 8: ChronosFinBERT Architecture.

4.2.4 TFT-FinBERT and TCN-FinBERT

Two multimodal fusion architectures pair a supervised temporal encoder (TFT or TCN-Transformer) with frozen FinBERT. In both variants, each news article associated with a given sample (up to 32 articles) is independently encoded by frozen FinBERT to produce a 768-dimensional [CLS] embedding. A four-head cross-article self-attention mechanism with a padding validity mask then pools these per-article embeddings into a single 768-dimensional news context vector, aggregating over variable numbers of daily articles without sensitivity to ordering or padding artifacts.

The temporal encoder produces a context vector from its final timestep: 128-dimensional for TFT and 256-dimensional for TCN-Transformer. Learned ticker embeddings (dimension 32) and GICS sector embeddings (dimension 16) supply static categorical context. The news context, temporal context, ticker embedding, and sector embedding are concatenated into a single joint representation, normalized via LayerNorm, compressed to 256 dimensions through a dense projection with ReLU activation and dropout, and mapped to a 5-step prediction by a linear head. The prediction represents a vector of returns relative to the last observed close. The loss during training is Huber [8] ($\delta = 1.0$).

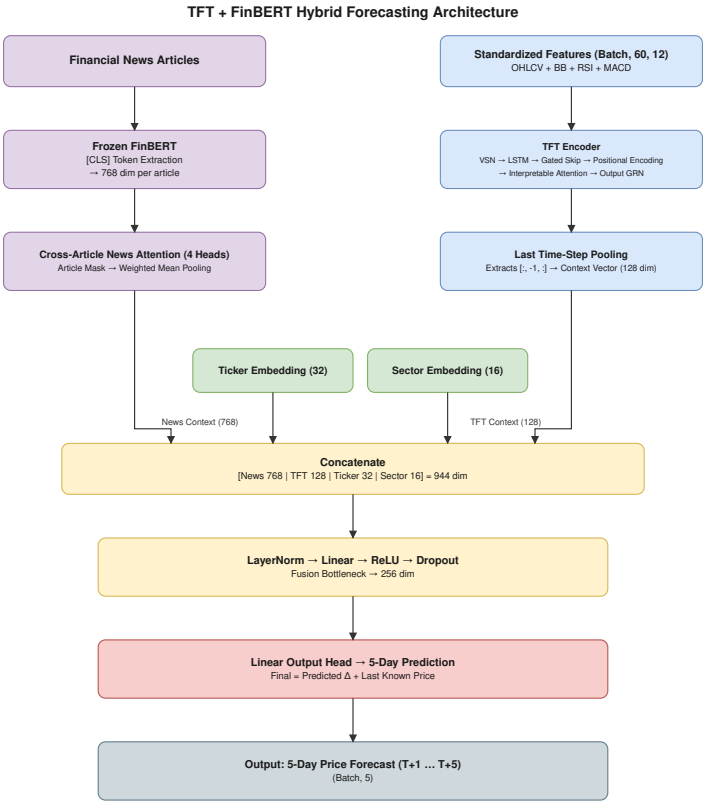


Figure 9: Late-fusion multimodal architecture combining TFT temporal features with FinBERT embeddings.

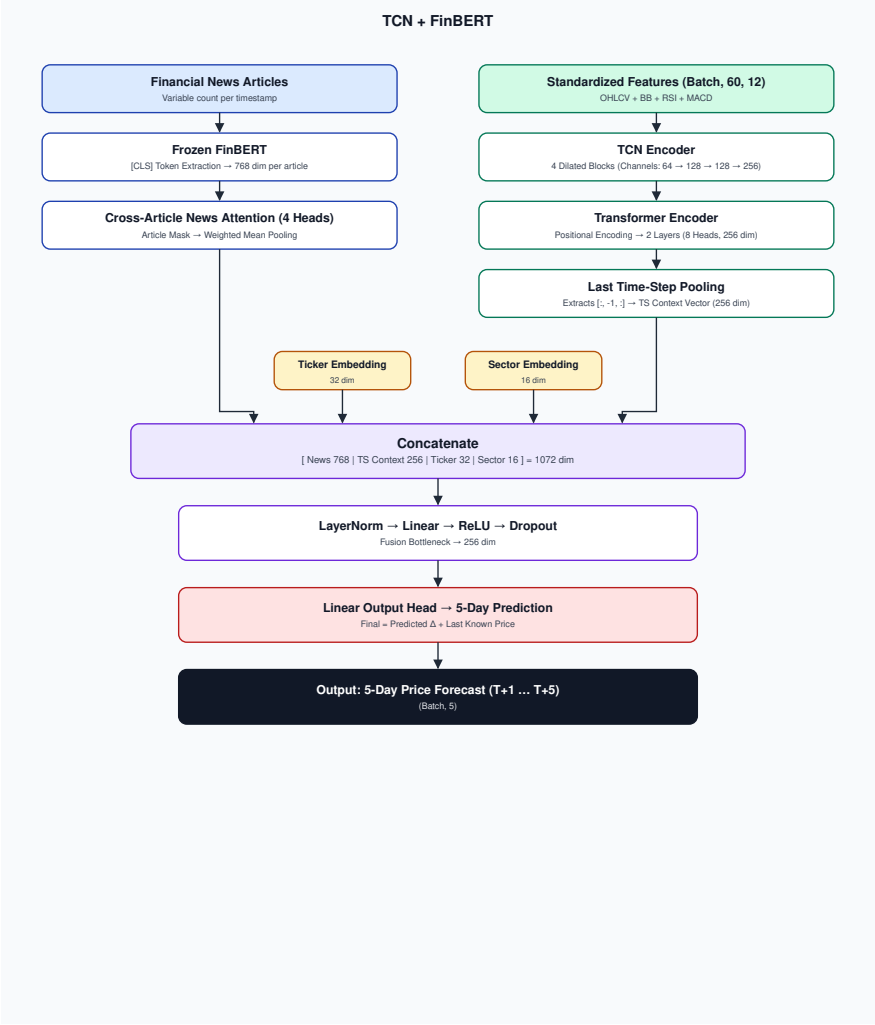


Figure 10: Late-fusion multimodal architecture combining TCN-Transformer temporal features with FinBERT embeddings.

4.2.5 FinBERT + CNN Price Encoder + Transformer Decoder

This architecture integrates price history and news through a full encoder-decoder design. On the news side, daily news articles are independently encoded by FinBERT with a per-day, per-article learnable attention weight to aggregate article embeddings within each trading day. Positional encodings are applied across the historical window to produce a time-indexed sequence of daily news representations of shape $[B, T, 768]$. On the price side, the full historical close price sequence is processed by a one-dimensional convolutional encoder comprising convolution, batch normalization, ReLU activation, and global average pooling, yielding a compact price representation of shape $[B, C_{\text{price}}]$. The price and news representations are projected to a common dimension and summed to form the encoder output.

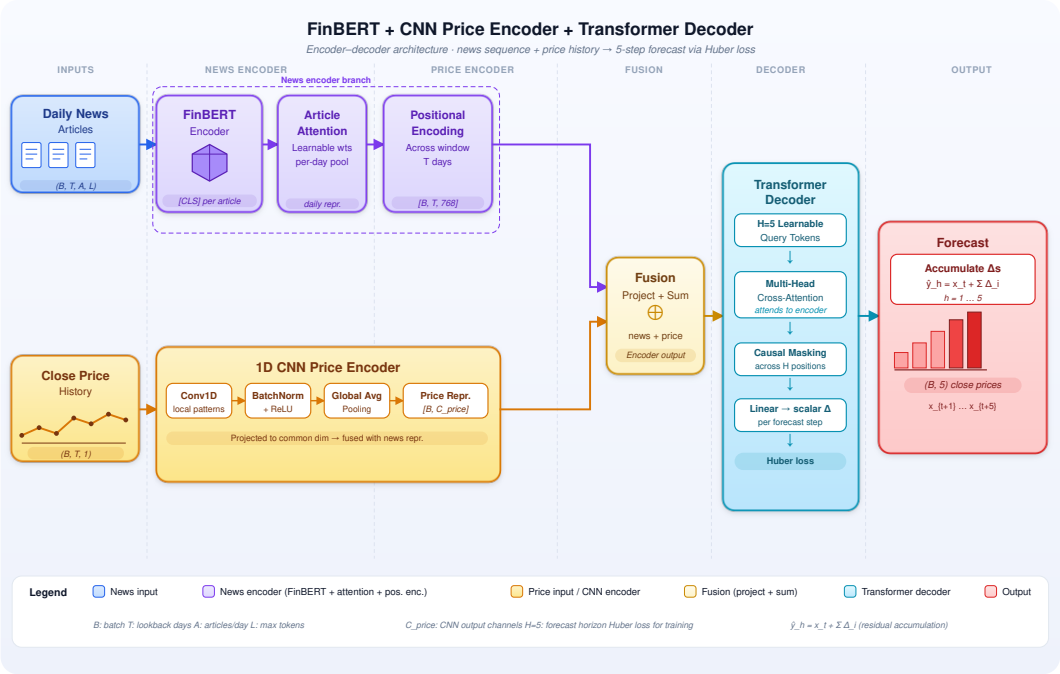


Figure 11: Finbert + CNN + Transformer Decoder architecture

A Transformer decoder receives $H = 5$ learnable query tokens, one per forecast step, and attends to the encoder output via standard multi-head cross-attention with causal masking across forecast positions. The decoder output at each position is projected to a scalar price increment; these increments are accumulated onto the last observed close x_t to reconstruct the predicted close price sequence. End-to-end training uses Huber loss [8].

Graph-based architectures were initially explored as a natural extension for the relational structure. A Temporal Graph Neural Network (TGNN) model was implemented and is available in the project repository, but resource and time constraints prevented full training and evaluation within the scope of this capstone. Incorporating graph-structured sector representations into the forecasting pipeline remains a concrete direction for future work.

5 Results

Model	RMSE	MSE	MAE
Time Series			
LSTM	0.03794	0.00143	0.02939
PatchTST	0.233	0.0543	0.1466
TSMixer	0.2228	0.0496	0.1474
Chronos-2	1.3505	1.8241	1.1954
Chronos T5-Large	1.3669	1.8684	1.2130
TFT	0.0681	0.00463	0.0358
TCN-Transformer	0.0673	0.00453	0.0355
Text Only			
FinBERT	0.1821	0.0332	0.1602
Time Series + News			
Chronos T5-FinBERT	0.1249	0.0156	0.0418
TCN-FinBERT	0.03029	0.00091	0.0286
TFT-FinBERT	0.038212	0.00146	0.03146
FinBERT+CNN+Transformer	0.0316	0.00099	0.0302

Table 2: Comparative performance summary across models

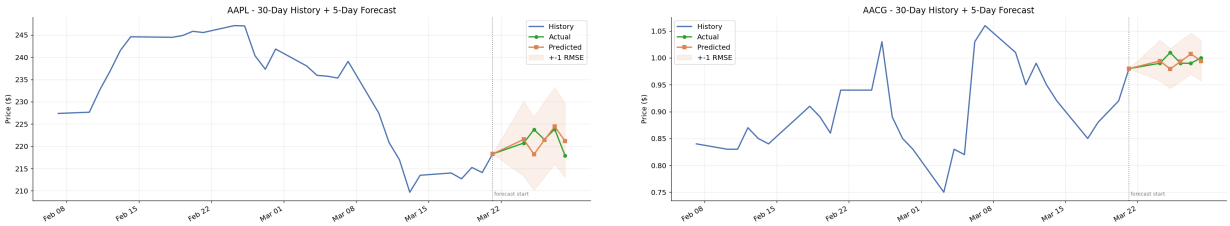


Figure 12: LSTM forecasts on example tickers.
The LSTM forecasts capture the broad price direction for both tickers, but they smooth out sharp local movements and lag when the true series changes abruptly.

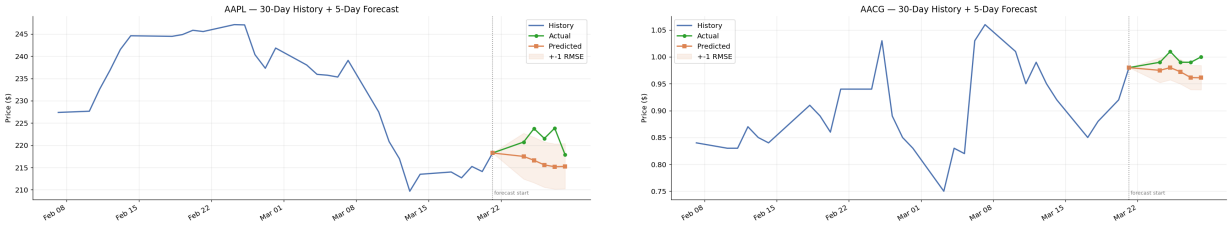


Figure 13: PatchTST forecasts on example tickers.
PatchTST should preserve short-term temporal patterns more effectively than the LSTM baseline but the forecasts still flatten in the five-day window.

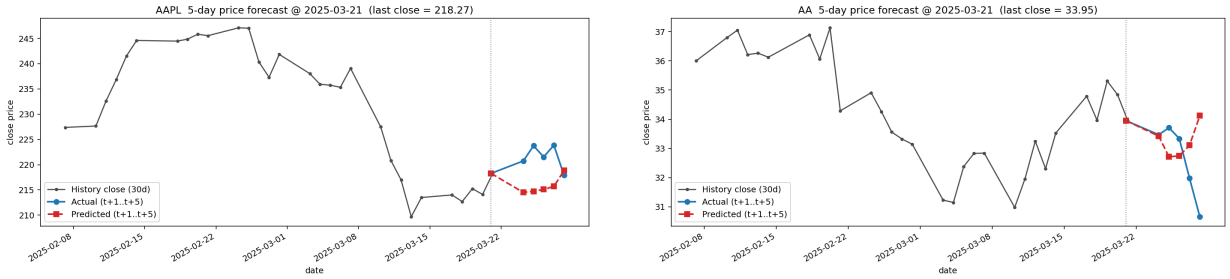


Figure 14: TSMixer forecasts on example tickers.
These examples show that TSMixer follows the overall level of each series, but its predictions deviate more on volatile segments, indicating weaker robustness to rapid regime changes.

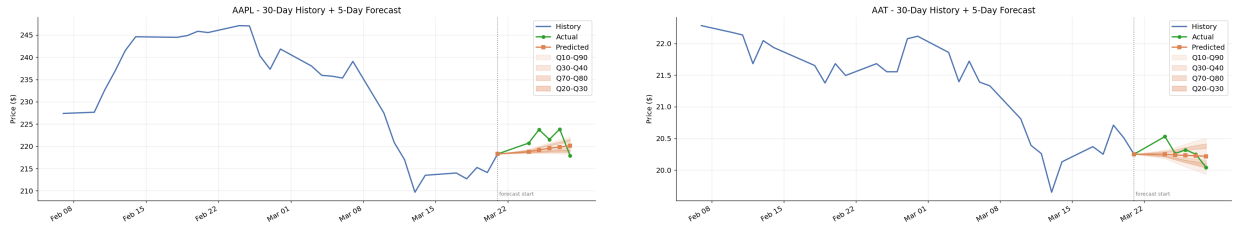


Figure 15: Chronos T5-Large forecasts on example tickers.

The Chronos T5-Large forecasts still tend to collapse toward the mean over the rollout horizon, but they generally capture the overall direction of the observed trend. The model produces multiple forecast quantiles, such as the 10th to 90th percentile range, which helps characterize uncertainty and provides a more informed basis for evaluating the prediction.

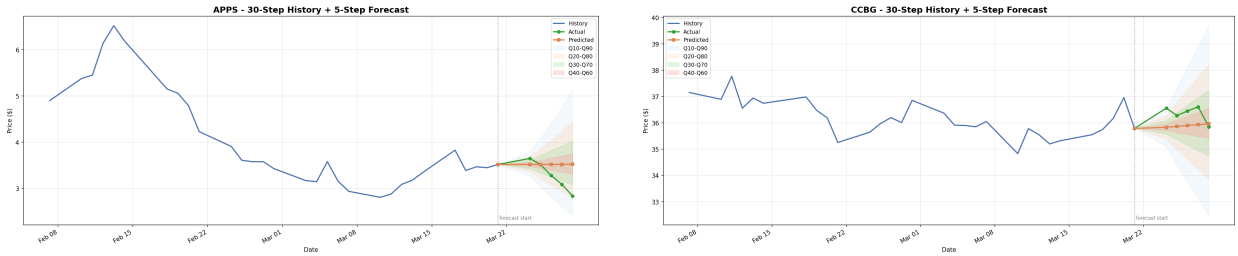


Figure 16: Chronos-2 forecasts on example tickers.

Chronos-2 also shows some mean-collapsing behavior in its multi-step predictions, but it often gets the broader price direction correct. By generating quantile forecasts, including ranges such as Q10–Q90, the model gives a useful uncertainty band that can guide how to interpret the forecast rather than relying on a single point estimate.

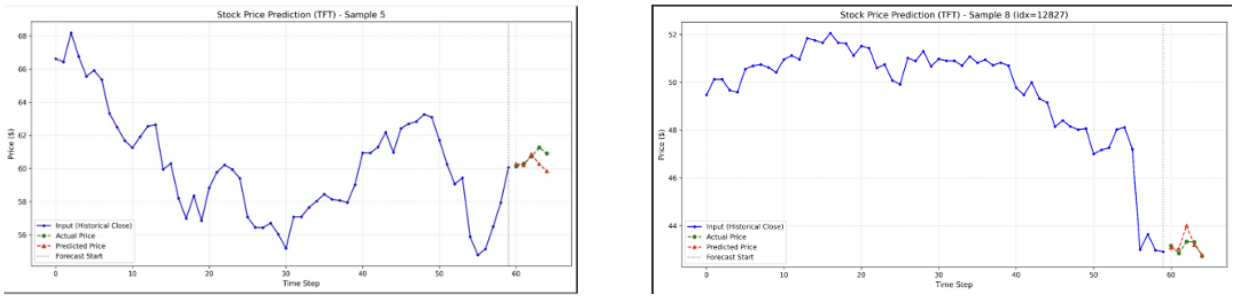


Figure 17: TFT price forecasting results on representative stocks.

The TFT examples illustrate stable multi-horizon forecasts with good trend tracking, although the model still underestimates some local peaks and troughs in the target series.

6 Discussion

6.1 Findings

Among the unimodal architectures, the two-layer stacked LSTM achieves the lowest RMSE, outperforming the more complex PatchTST, TSMixer and other models. Since all models receive identical engineered features — log returns, rolling return statistics, RSI(14), ATR(14), MACD, signal line, and Bollinger Bands — this gap reflects genuine architectural differences rather than information asymmetry. The LSTM's advantage over PatchTST and TSMixer most plausibly stems from its sequential inductive bias. LSTM aligns naturally with the short-horizon forecasting task, where recent price behaviour and indicator readings carry disproportionate predictive signal.

PatchTST and TSMixer impose design assumptions that actively conflict with the structure of these engineered features. PatchTST applies RevIN normalisation across the full lookback window before patching. For bounded indicators such as RSI and MACD, this introduces a redundant re-standardisation that can suppress information these indicators are designed to carry. Its patch design also misaligns with the dominant five-day weekly cycle, potentially averaging over event-driven moves that should be treated as discrete change points. TFT and TCN-Transformer perform more competitively because their respective Variable Selection Networks and dilated causal convolutions are better suited to the heterogeneous temporal structure of the feature set.

The zero-shot foundation models perform worst overall. Although pre-trained on a large and diverse collection of datasets, their temporal priors appear poorly matched to equity series, which are characterised by structural breaks and regime change correlation. Fine-tuning the Chronos encoder on task-specific data and pairing it with FinBERT substantially closes the gap: ChronosFinBERT achieves approximately a ten-fold improvement over zero-shot Chronos. This confirms that the Chronos architecture retains useful temporal inductive biases that can be unlocked through targeted adaptation, but ChronosFinBERT still trails the best supervised multimodal models by a substantial margin, indicating that the pre-training regime of current time-series foundation models remains misaligned with the demands of financial forecasting.

The text-only FinBERT baseline confirms that news sentiment without price history is insufficient for reliable multi-step forecasting, functioning as a lower bound on the news modality in isolation. However, when news representations are integrated with a competent temporal encoder, consistent and substantial improvements emerge across all three fusion architectures: TCN-FinBERT, TFT-FinBERT and FinBERT+CNN+Transformer. These gains are consistent with the hypothesis that financial news provides temporally localised event signals — earnings surprises or analyst revisions — that are not recoverable from price history alone.

Comparing TFT-FinBERT and TCN-FinBERT reveals something interesting. The two models are built almost identically in terms of how they combine news and price data, and they perform similarly when used without news at all. Yet once news is added, one pulls noticeably ahead of the other. This suggests that the underlying price model matters a lot — a stronger price encoder is simply better at making use of the news signal fed into it. Both models also benefit from an attention step that reads multiple news articles and decides which ones are most relevant for a given day, rather than just averaging them all together. This selective reading appears to be an important

part of why the news modality helps at all. FinBERT+CNN+Transformer works differently. Rather than processing price and news separately and then stitching them together at the end, it blends them from the start and lets each forecast step actively look back and pull whatever information seems most useful — whether that's a price pattern from several weeks ago or a specific news event. This makes it one of the best-performing models in the study. Taken together, these results support the conclusion that fusion mechanism quality meaningfully mediates the value of the news modality, and that architectures permitting dynamic interaction between temporal and textual representations consistently outperform those that naively pool and concatenate the two modalities.

6.2 Limitations

Several limitations qualify the conclusions drawn from this study. First, while all unimodal models receive the similar engineered feature sets, the architectural hyperparameters — patch size, lookback length, hidden dimensions, and loss functions — were not jointly optimised across models under a single shared protocol. Second, all evaluations are conducted on a five-day horizon with a fixed 70/15/15 temporal split. Third, the structured financial reports modality (SEC XBRL data) was preprocessed and forward-filled but was not incorporated as a modality in any of the evaluated fusion architectures; its contribution to predictive accuracy remains unexamined in the experimental protocol. Fourth, the Temporal Graph Neural Network architecture, which was designed to exploit sector-level relational structure, could not be fully evaluated within the timeline, leaving a potentially impactful architectural variant unevaluated.

6.3 Future Work

Several directions extend naturally from these findings. The most immediate is the systematic ablation of fusion components within a single held-out architecture: replacing cross-article self-attention with mean pooling, replacing the learned query tokens in the decoder with fixed positional tokens, and varying the number of fine-tuned FinBERT layers, would each isolate specific design choices whose individual contributions are currently entangled.

A second direction is the integration of the SEC XBRL financial report modality as an input to the fusion architectures evaluated here. Structured fundamental data (cash flow, equity, earnings metrics) operates at a lower frequency than news and price history, but its periodic disclosure events are known to induce persistent price adjustments that extend beyond the five-day forecast horizon. Incorporating it via a separate low-frequency encoder with learned temporal alignment to daily observations is a concrete next step.

Third, the sector-relational TGNN architecture deserves full evaluation. Industry-level co-movement is a well-documented feature of equity markets, and graph convolution over a sector adjacency matrix could capture cross-stock dependencies that no single-stock model can represent.

Fourth, the evaluation horizon and split protocol could be extended to a rolling-window or walk-forward scheme, which would provide a more realistic assessment of model performance over market regimes that were not observed during training, including the non-stationary structural breaks that

proved problematic for the Chronos foundation models.

Finally, this study evaluates regression accuracy via RMSE, MSE, and MAE. Future work should augment the quantitative evaluation with directional accuracy metrics, Sharpe ratio simulations, and maximum drawdown analysis under a simple signal-following strategy.

7 Conclusion

This report presented a systematic benchmarking study of unimodal and multimodal architectures for short-horizon stock price forecasting on S&P 500 constituents, using the FinMultiTime dataset as a unified data source across modalities. Our principal finding is that the incorporation of financial news via FinBERT embeddings yields consistent and substantial improvements over all unimodal temporal baselines. TCN-FinBERT achieves the best overall performance, representing a significant reduction in metrics relative to its unimodal counterpart and outperforming all other models evaluated. The encoder-decoder FinBERT+CNN+Transformer architecture achieves comparable accuracy with additional interpretability advantages arising from its per-step cross-attention maps. These results establish that event-driven textual signals carry information about short-horizon price dynamics that is not recoverable from price history alone within practically feasible lookback windows.

A second finding is that architectural fit to the financial domain matters independently of input representation. PatchTST and TSMixer, evaluated on the same engineered feature set as all other unimodal models, underperform substantially. Their degradation is attributable to structural mismatches: PatchTST's RevIN distorts bounded technical indicators and channel independence discards cross-indicator semantics, and TSMixer's symmetric time-feature mixing lacks ability to handle irregular regime switching in equity returns. These results caution against direct transfer of architectures validated on general-purpose benchmarks to financial forecasting tasks.

A third finding is that the fusion mechanism quality mediates how much of the textual signal is ultimately exploited. Architectures that allow dynamic interaction between temporal and textual representations — through cross-article self-attention pooling or through a learned decoder cross-attention — outperform naive concatenation approaches, even when the underlying temporal encoders are of comparable unimodal quality.

Another finding is related to the failure modes of general-purpose pre-trained models. Zero-shot Chronos foundation models produce dramatically worse forecasts than any supervised baseline, confirming that pre-training on broad non-financial corpora does not generalise well to the non-stationary, regime-switching dynamics of equity markets. ChronosFinBERT partially recovers through fine-tuning, but still trails all fully supervised multimodal models, underscoring that domain adaptation remains a prerequisite for deploying time-series foundation models in financial settings.

Collectively, these findings offer practical guidance for the design of multimodal financial forecasting systems. Domain-specific supervised training, rich engineered temporal features, and attention-based fusion mechanisms are each necessary components of a performant pipeline; no single factor alone is sufficient. The structured financial report modality and graph-based sector representations remain promising directions that were not fully evaluated in this study, and their integration into the framework described here constitutes a concrete and well-motivated agenda for future research.

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